Eco 213: Basic Data Analysis and Introductory Econometrics

Department of Economics, Shiv Nadar University

Monsoon 2020

Instructor: Gitanjali Sen (Gitanjali.sen@snu.edu.in)

TA: Sarthak Sharma (email: ss134@snu.edu.in)

Class Hours: Wed and Fri: 5 pm to 6:25 pm (Meetings online)

Credits: Credits (L:T:P)- 3:1:0

This is a compulsory course for Economics minor degree. Econ major students should not register in this course.

Since the complete semester is taught online due to the pandemic, a well-effective teaching-learning process requires serious effort and collaborations from both students and teachers. The starting point of that being regular attendance in the class, I would request students to follow my classes on a regular basis, attend lectures, complete all required readings/text books, assignments on given time. If you miss any class due to any connectivity issues, please try to make up for that by using the recorded lectures, but don't make missing lectures a habit, because the recorded lectures are not necessarily interactive in nature.

OBJECTIVES AND PRACTICE

1. Course Summary

This course introduces the basics of the practice of modern econometric techniques. A detailed discussion of basic ideas in probability and statistics and the linear regression model will be presented. The topics included in the course are: the simple regression model, multiple regression models, classical assumptions about disturbances, hypothesis testing, violation of classical assumptions, multicollinearity, heteroskedasticity, omitted variable bias, functional forms, dummy variables, and goodness of fit. This course will be intensive in assignments both analytical and data oriented. To complete some assignments and the project the students will also be introduced to STATA, statistical analysis software.

2. Course Aims

1. Deepen mathematical and statistical skills
2. Foster analytical and critical thinking
3. Develop problem solving abilities
4. Enhance learning by doing.

5. Formulation and specification of econometric models. The economic models are formulated in an empirically testable form.
7. Estimation and testing of models
8. Use of econometrics methods and models to develop analytical and decision-making abilities through training in quantitative and model-based methods of analysis.

LEARNING OUTCOMES

On successful completion of the course, students will be able to:
1. Demonstrate an appreciation of the relationship between theoretical statistics and the econometric techniques used to analyze real world data.
2. Apply the mathematical techniques of probability and distributions to inference, estimation and hypothesis testing, which are the formal methods by which we learn from noisy data, random samples, and other such uncertain real-world measurements.
3. Demonstrate problem solving skills, especially how to analytically prove results from a set of given assumptions.
4. Demonstrate effective communication of the analytical complexity of the problems encountered.
5. They should be able to use a statistical/econometric computer package (STATA) to estimate an econometric model and be able to report the results of their work in a lucid manner

CURRICULUM CONTENT

Course Outline

I. Introduction and recap of basic ideas in probability and statistics, notion of causality in econometric analysis.

Appendices A, B and C; Chapter 1 (Woolridge)

1.1. Probability Distribution and Densities
1.2. Mathematical Expectations
1.3. Some Useful Distributions
Normal Distribution. Chi Square Distribution. t-Distribution. F Distribution.
1.4. Estimation
Method of Moments Estimation. Unbiasedness. Efficiency. Best Linear Unbiased Estimator.
1.5. Hypothesis Testing

II. The simple regression model, Ordinary Least Squares, multiple regression analysis – estimation and inference, hypothesis testing and confidence intervals, issues of multicollinearity
Chapters 2, 3 and 4 (Woolridge)

III. Large sample theory, basics of OLS asymptotic theory
Appendix C; Chapter 5 (Woolridge)

IV. Beta coefficients, functional forms, measures of goodness of fit, residual analysis, heteroskedasticity and robust inference
Chapters 6[6.1, 6.2, 6.3], 8[8.1, 8.2, 8.3] (Woolridge)
[Some components in this unit will be optional and decided towards the end based on the course requirements and progress of the class]

V. Analysis with qualitative information –binary or dummy variables, use of categorical variables, linear probability models
Chapter7(Woolridge)

The course will encourage students to learn Stata, although it is not expected from the students to come with a prior Stata knowledge. But due to the online delivery mode in this semester, and with due consideration of students' inability to work and learn in groups, the instructor and the TA will try to facilitate a few options for learning Stata. The end term project will involve basic knowledge of Stata and application of that in Econometrics. Therefore, it is imperative that student follow the classes from the beginning and learn the requisite skills as and when asked by the Instructor/TA.

TEXT BOOK:

1 **Introductory Econometrics: A Modern Approach"** by Jeffery M Wooldridge (5th Edition).
2 Introduction to Econometrics by Christopher Dougherty (fourth Edition)
3 For STATA there are some very useful resources available online – http://www.ats.ucla.edu/stat/stata/, http://data.princeton.edu/stata/

ASSESSMENT:

Component (out of Total 100 marks)	Total marks and Weights
In-class or at home assignments / quizzes and random viva based on the assignment/quizzes (a Total of 3, with each having weight of 20 points)	Total: 60 (3 x 20 marks))
End term project using Stata knowledge (could be divided into a few smaller components	Total: 40

Plagiarism should be strictly avoided. Any sign of plagiarism found in course assignments/quizzes or project will be taken very seriously, which may lead to getting a fail grade in the course. If you need any help in understanding the material, you should approach the TA or the instructor.

All deadlines should be strictly followed. Please note that submission deadlines are the last date, and you are encouraged to submit a work before the deadline, to avoid any last minute hassles/problems at your end. Please do not ask for any extension of the deadline or timing of submission of any work. You may be asked question randomly in a class based on your submitted assignments, which will be used to grade your assignment jointly along with your submitted assignment. Since you are not taking any exam at physical class room gathering, viva (oral exam) may play a significant role in judging your written answers and related understanding of any particular assignment.